

Role of Prenatal Exposure to Inflammatory Stress on Microglia Density and Amyloidogenesis

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Background

Alzheimer's disease (AD) is a progressive neurodegenerative disease marked by memory loss, cognitive decline, and behavioral deficits. It is estimated that an average of 5.4 million Americans currently have Alzheimer's disease (Alzheimer's Association, 2016). Within the next 30 years, that number is expected to grow to over 13 million people afflicted with AD—there is currently no known cure.

Several variables contribute to AD pathophysiology that ultimately lead to clinical manifestations of the disease. Amyloid- β (A β) protein dysfunction and aggregation is one hallmark of AD—A β is a product of aberrant amyloid precursor protein cleavage. A β plaque area was used in this study as a marker of AD onset/progression.

Microglia are the primary immune cells of the brain, and are responsible for phagocytosis of pathogens, toxins and cellular debris. Under normal homeostatic conditions, microglia survey their microenvironment for abnormalities in the extracellular space. In AD, microglia undergo gliosis and shift into an ameboid phenotype capable of phagocytosis of A β monomers, oligomers and plaques. These morphological changes in microglia are also accompanied by a pro-inflammatory cytokine profile.

Priming of microglia is likely one important aspect of microglia facilitated removal of A β and the chronic inflammatory state found in AD. Priming occurs when immune cells experience a stressor that produces an activated state—this activated state alters the gene expression signature of the cell, leading to a heightened response to subsequent stimuli (Holtman et al., 2015).

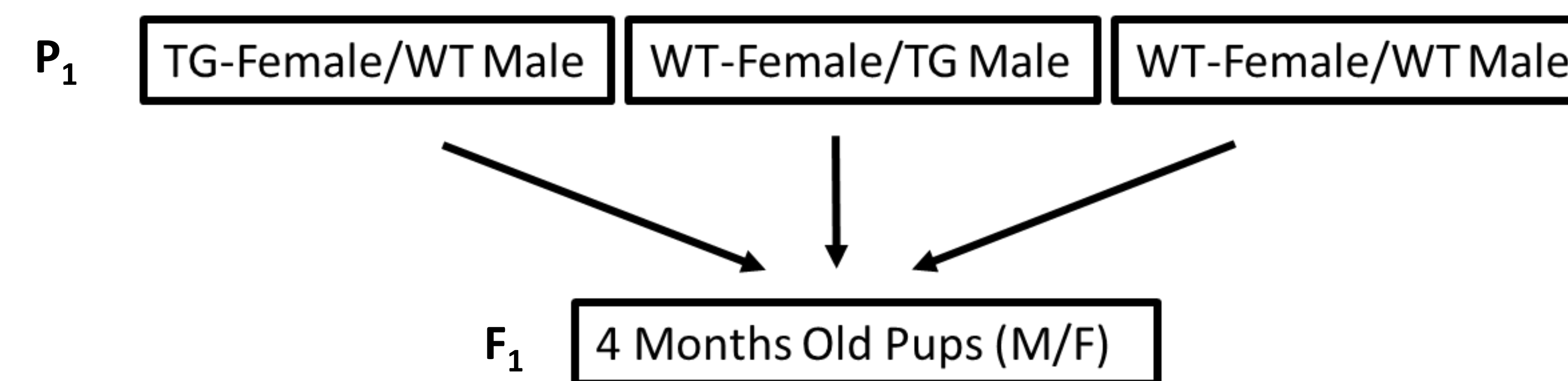
In the United States, females make up over 60% of the AD demographic (Mufson, 2016). Women also show a faster rate of progression of clinical symptoms, as measured by psychometric test batteries and Braak staging (Sperling, 2011). Sex differences may provide insight into the mechanisms involved in the relationship between neuroinflammation and AD pathophysiology.

Hypothesis-Early exposure to inflammatory stress worsens AD onset and progression, specifically in females compared to males.

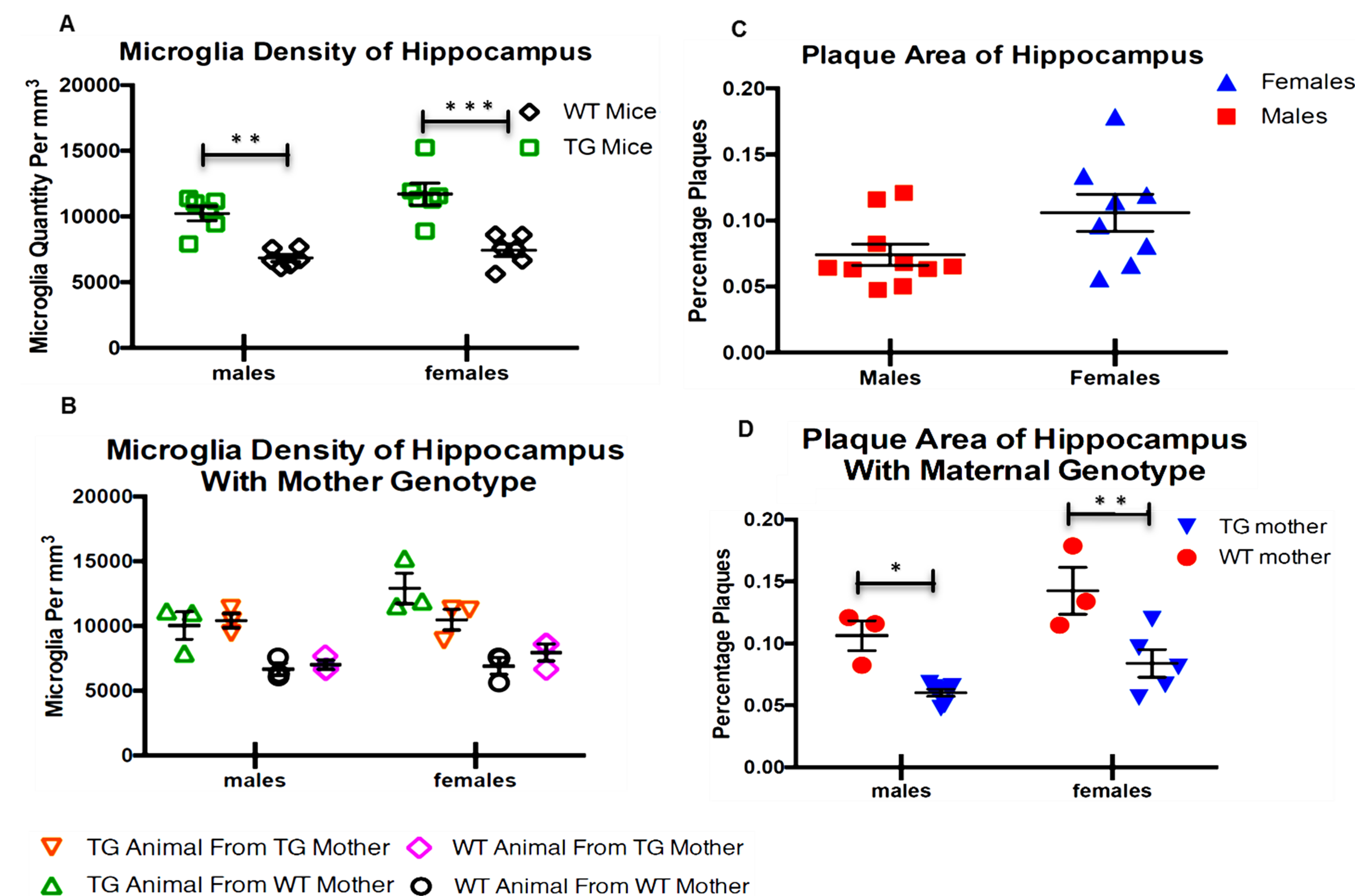
Aim 1- Determine the effect of exposure to prenatal inflammatory stress in AD onset and progression.

Aim 2-Determine if sex differences are present with respect to amyloidogenesis and microgliosis.

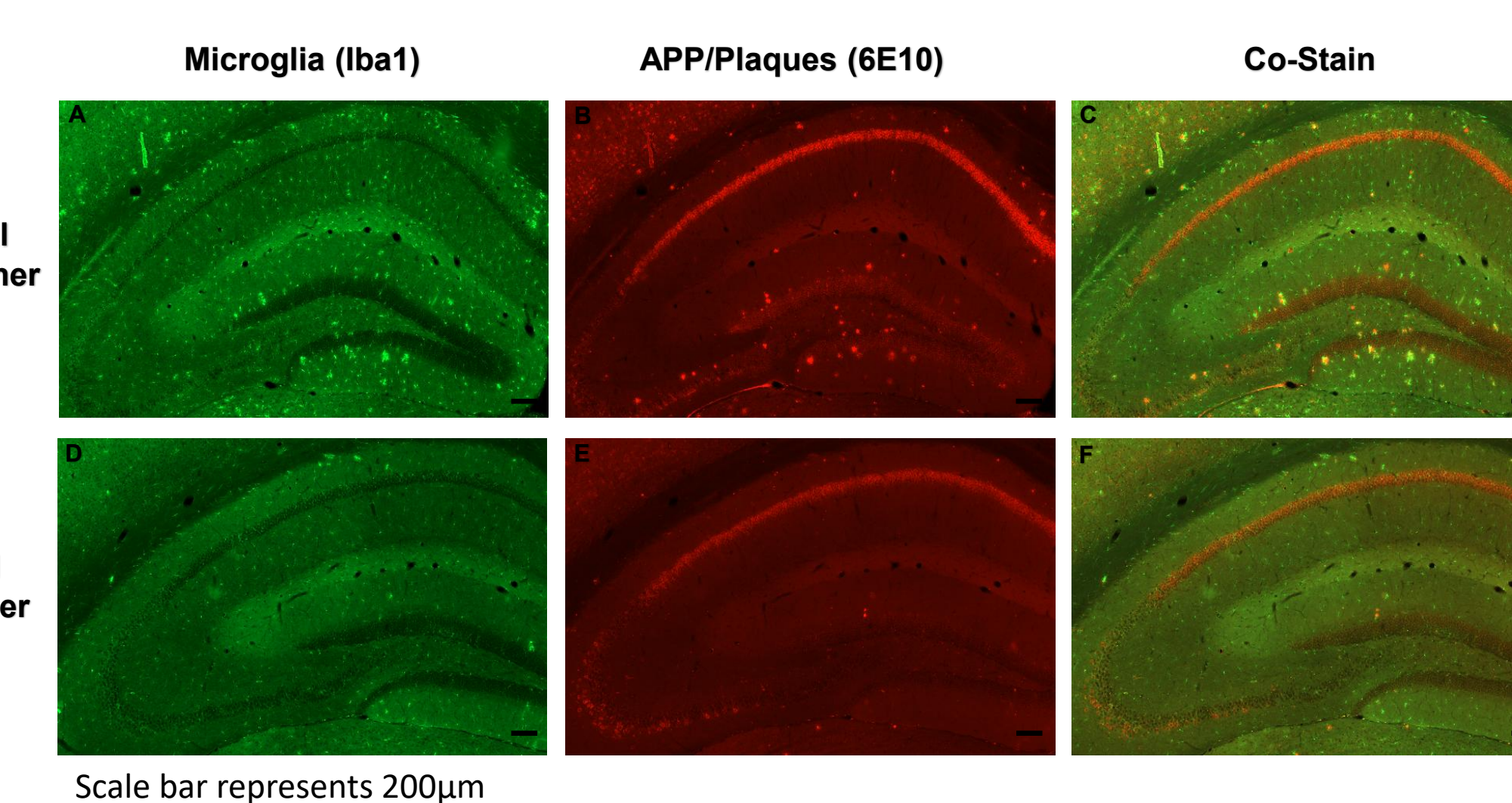
Experimental Design



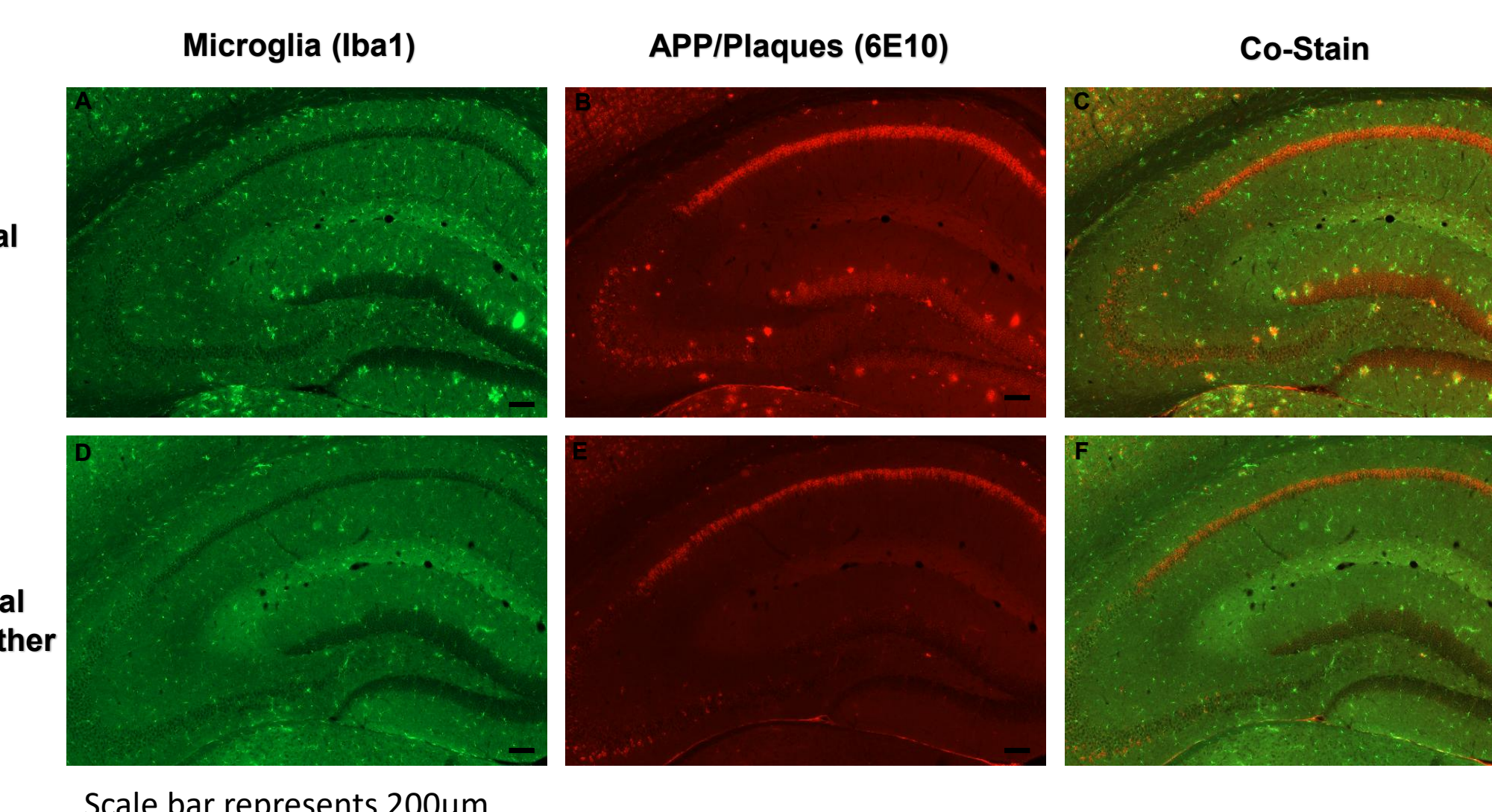
Hippocampal Microglia Density and Plaque Area



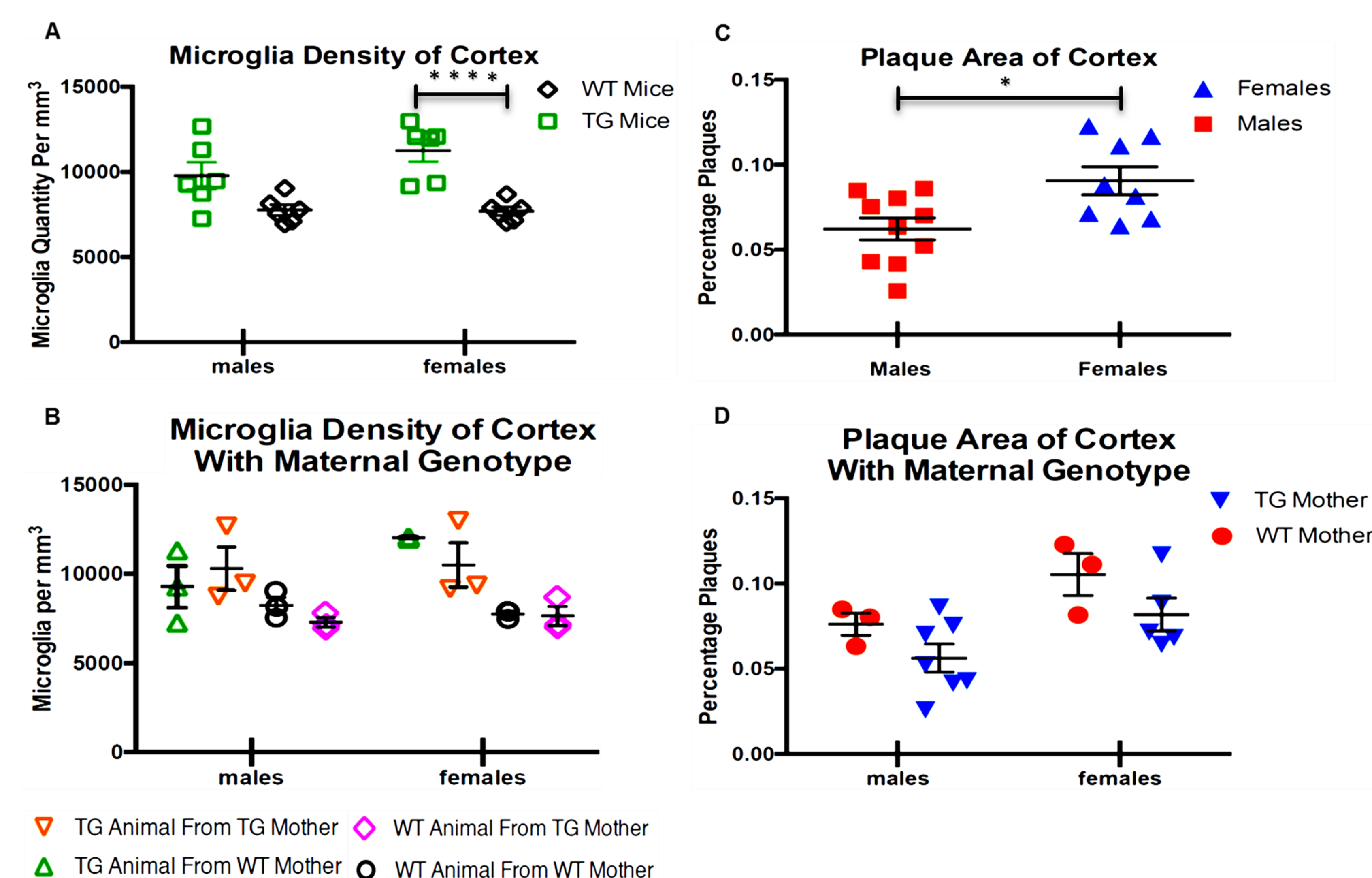
Male Hippocampal Microglia Density and Plaque Area



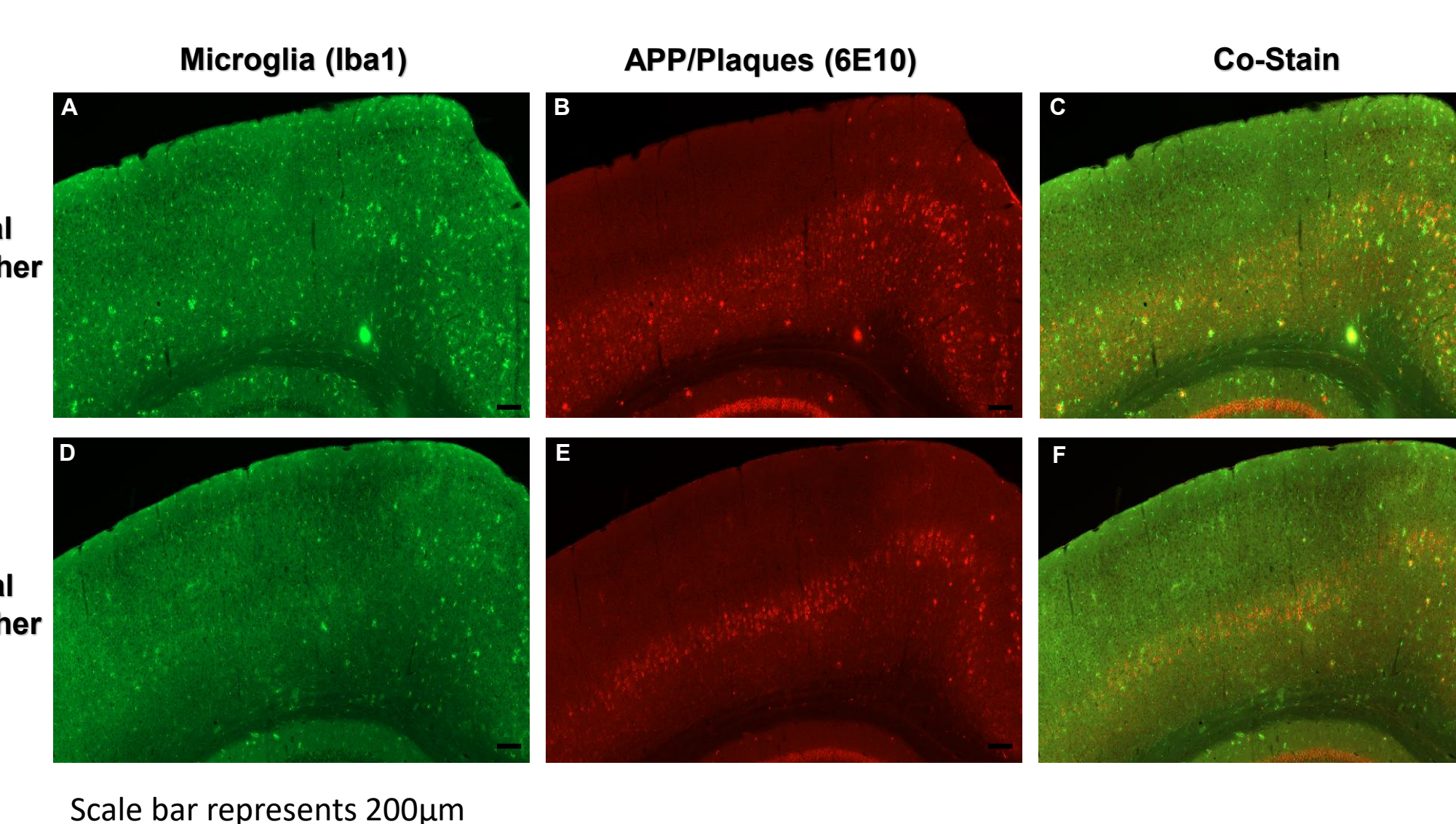
Female Hippocampal Microglia Density and Plaque Area



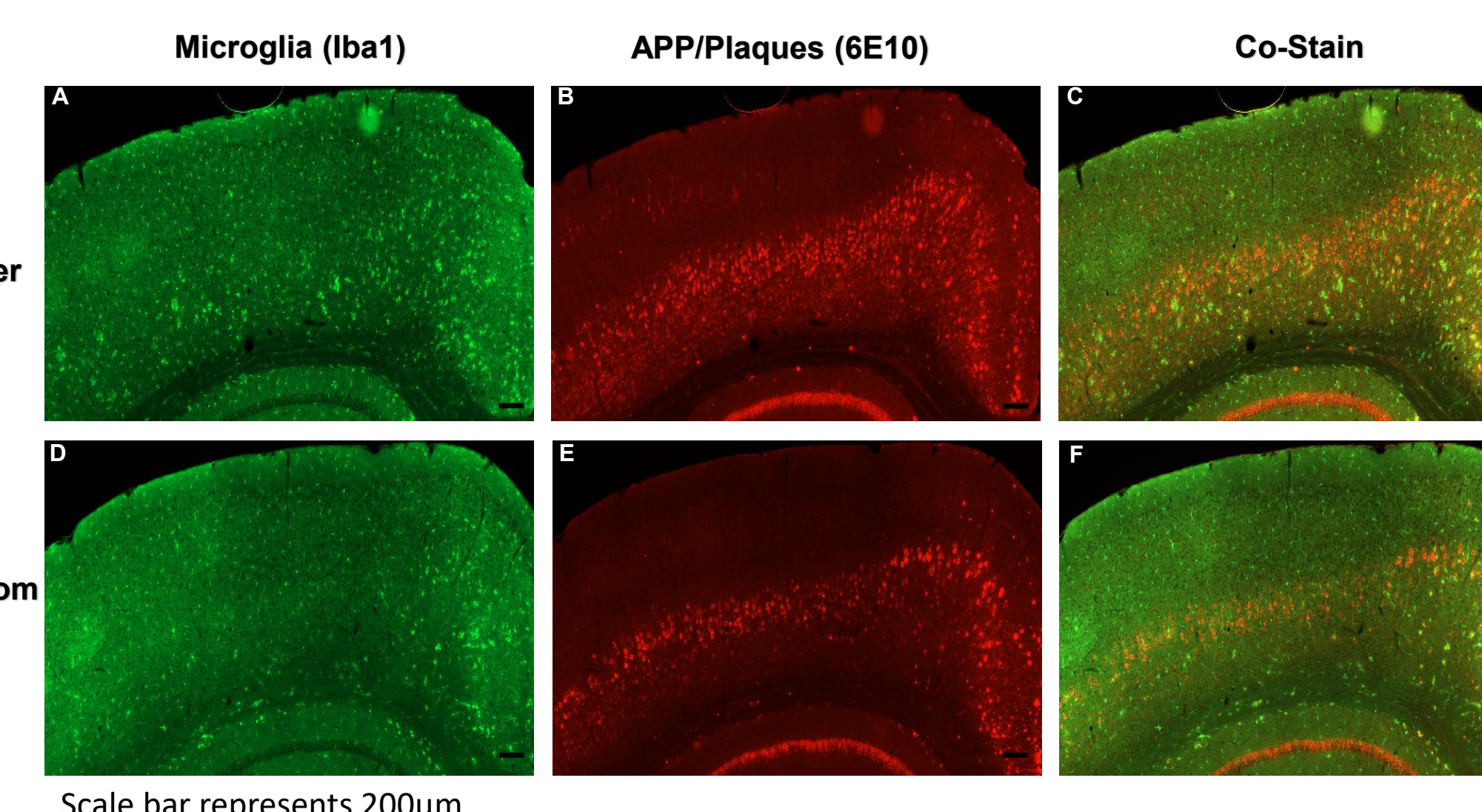
Cortical Microglia Density and Plaque Area



Male Cortical Microglia Density and Plaque Area



Female Cortical Microglia Density and Plaque Area



Material and Methods

The 5 X's Familial Alzheimer's disease (5xFAD) transgenic (TG) mouse model was used in this study. This model incorporates 3 human amyloid precursor protein (APP) mutations and 2 human presenilin 1 (PSEN1) mutations that produce A β dysfunction and deposition. This model reliably produces plaques by 2 months of age. The mice used in this experiment were euthanized for tissue analysis at 4 months of age.

TG mice display a marked chronic pro-inflammatory state compared to wild type mice. Rodent models mimicking prenatal inflammatory insult establish a clear transfer of inflammatory cytokines to fetal brains (Cai et al., 2000; Paintiliaet et al., 2004; Liverman et al., 2006). Because of this phenomenon, mice born from TG mothers will be exposed to prenatal inflammatory stress. This inflammation occurs before the activation of the human mutations in the 5xFAD model—before onset of amyloidogenesis.

Histological analysis using 30µm vibratome sections was conducted for identification of APP/A β (6E10) expression in the hippocampus and surrounding cortical regions, as well as microglia (Iba1) in this area. Unbiased stereology software was used to quantify plaque area and microglia density.

Discussion

Male and female mice exposed to prenatal inflammatory stress showed less plaque load compared to mice not exposed to prenatal immune stress, in both the cortex and hippocampus. Males exposed to inflammatory stress showed 46% less plaque area compared to animals not exposed to prenatal immune stress, while females showed 59% less plaque area in animals exposed to prenatal immune stress.

These results suggest that reduced amyloidogenesis and/or removal of A β is occurring in animals that are exposed to early inflammatory stressors. This may suggest a priming effect of microglia, where prior inflammation increased the effectiveness of plaque removal. However, this was not correlated with an increase in microglia density, as no significant difference between exposure to prenatal stress and microglia density occurred.

A sex-difference was observed for plaque load in cortical regions, with females displaying greater plaque area. This phenomenon was also observed in the hippocampus but was not deemed statistically significant ($P=.055$). No sex-differences were observed between animals exposed to prenatal inflammation, suggesting that the influence of prenatal inflammatory stress did not significantly differ in females compared to males.